

INSTRUCTIONS FOR USING THE PROGRAM `simulate-polymer.bin`

Install Visual Studio Code on LINUX.

Install the VMD molecular dynamics visualization program on LINUX.

Copy the entire Program directory to your computer.

Look at the first line of the `commands.txt` text file:

```
./simulate-polymer.bin Simulations/Test4
```

To launch the program, type the following on the command line:

```
./simulate-polymer.bin Simulations/Test4
```

To execute without errors:

If you enter a different number of monomers (N) in the monomer positions file than the one in the `parameters.dat` file: error

If the positions file has monomers that are too close together: error (approximate distance between consecutive monomers: on the order of 1)

If the positions file has monomers that are too far apart: the molecule is split.

If the positions file name in `parameter.dat` does not match any `xxxxx.gro` file in the Program directory: error.

If the positions file is not sorted correctly: error

THE ENTIRE EXECUTION DEPENDS ON WHAT IS MODIFIED IN THE `parameters.dat` FILE:

Example of a `parameters.dat` file: (modify ONLY the numbers in the first 8 lines; subsequent comments can be modified or deleted)

```
T 0.0
eta .5
N 26
f_pull 2
IC_type k
meas_freq 1
sim_time 50
file_to_load K_3_L_tzero.gro
// This file (file_to_load.gro) must be in the
// folder where the .gro and .trr files are written

// T is Kelvin temperature (better T=0.0, but you are free to change it...)
// eta is the viscosity (a little bit of viscosity improves the behavior
// of the program)
// The value N (number of particles)
// f_pull is pulling forces on the ends of the polymer (force -x on the
// first monomer, and force +x on the last monomer).
// IC_type: r => random, k => knot
// meas_freq: number of recorded pictures: 1 for each iteration, 0.5 for
// each 2, etc.
// sim_time: number of iterations
```

The program is designed using the directive function of the `parameters.dat` file. It's not an intuitive interface, but that's how it was programmed. Sorry.

To visualize the polymer's motion, use the VMD program by running something like this on the command line (see commands.txt file):

```
vmd -e visualize-polymer.tcl -args Simulations/Test4/initial-condition.gro  
Simulations/Test4/trajectory-file-0.trr (this will display the iterations  
you specified in the sim_time parameter).
```

To see more iterations, you can run simulate-polymer.bin again, and to view them, use the same command ending in -file-1.trr, and so on.

The file analyse-polymer.bin executed in the console says you the knot type till K_10_165 and other calculations.

Good luck.